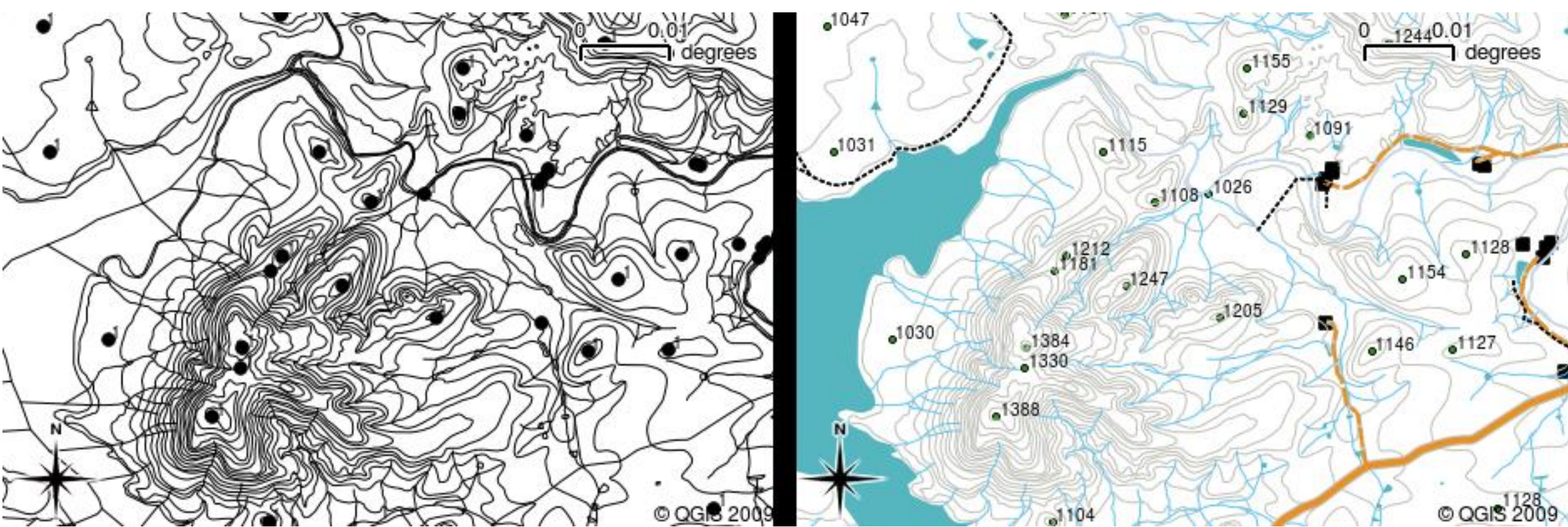


Vector Attribute Data

Objectives:	In this topic we describe how attribute data are associated with vector features and can be used to symbolise data.
Keywords:	Attribute, database, fields, data, vector, symbology

Vector Attribute Data

- If every line on a map was the same color, width, thickness, and had the same label, it would be very hard to make out what was going on.
- The map would also give us very little information.
- Take a look at Fig.1. for example.



- *Fig.1. Maps come to life when color and different symbols are used to help you to tell one type of feature from the next.*
- *Can you tell the difference between rivers, roads and contours using the map on the left?*
- *Using the map on the right it is much easier to see the different features.*

Overview

- In this topic we will look at how attribute data can help us to make interesting and informative maps.
- In the previous topic on vector data, we briefly explained that **attribute data** are used to **describe vector features**.
- Take a look at the house pictures in **Fig.2**.

Roof Colour: Red
Has balcony?: Yes
Year Built: 2000



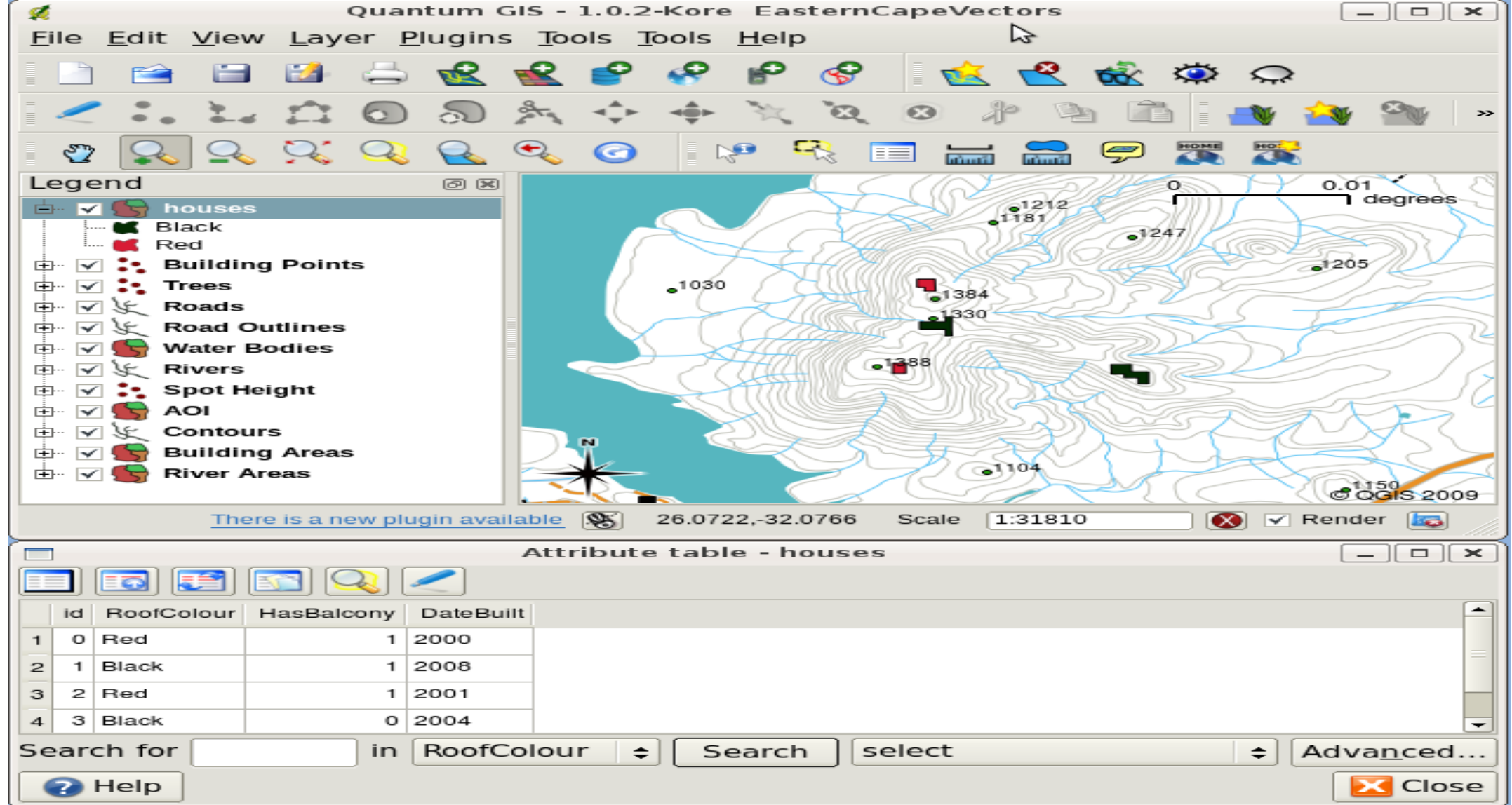
Roof Colour: Black
Has balcony?: No
Year Built: 2002



Fig. 2. Every *feature has characteristics* that we can describe. These can be visible things, or things we know about the feature (e.g. year built).

Vector Attribute Data

- The geometry of these house features is a polygon (based on the floor plan of the house), the **attributes** we have recorded are **roof color**, whether there is **a balcony**, and the **year the house was built**. 
- Note that attributes don't have to be visible things — they can describe things we know about the feature such as the year it was built.
- In a GIS Application, we can represent this feature type in **a houses polygon layer**, and the attributes in an **attribute table** (see [Fig. 4.8](#)).

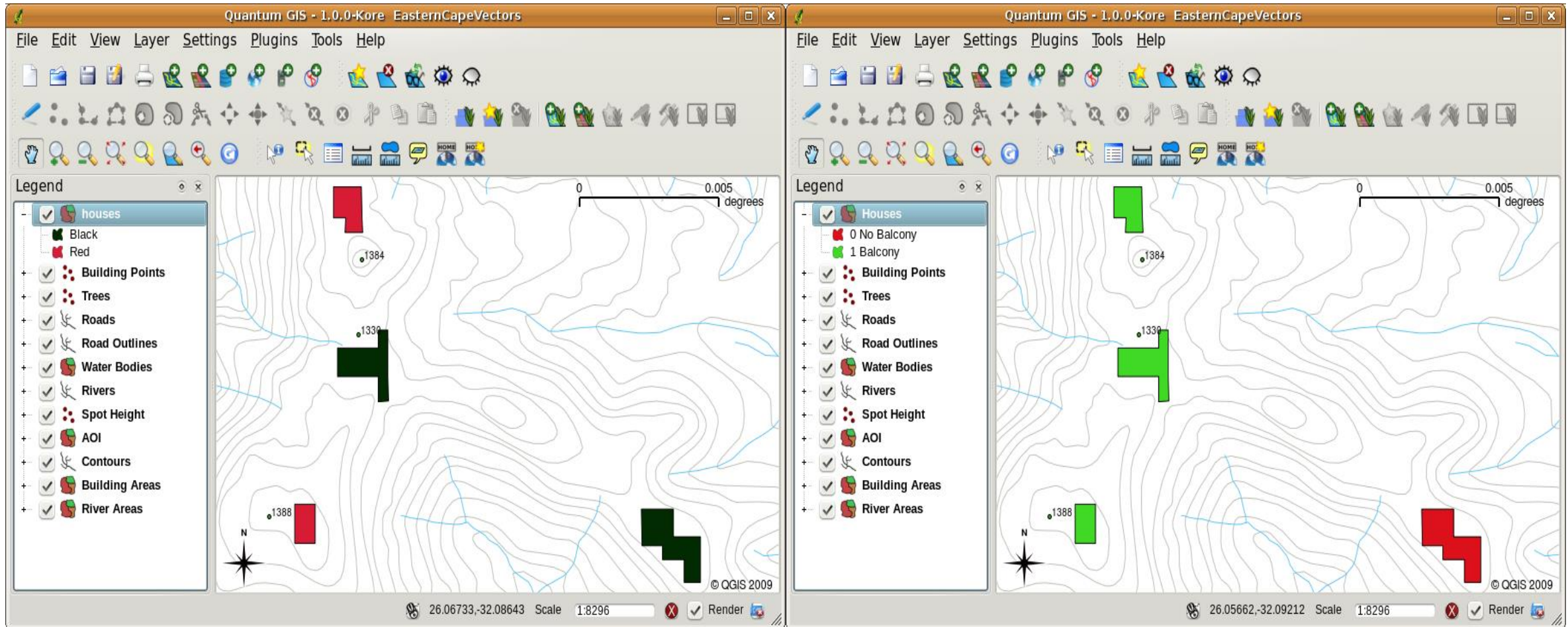


- A houses layer. *House features have attributes* that describe the houses' *roof colour* and other properties. The attribute table (lower image) lists the attributes for the house areas shown on the map.
- When a feature is highlighted in the table, it will appear as a yellow polygon on the map.

symbolology

- The fact that features have attributes as well geometry in a GIS Application opens up many possibilities.
- **For example** we can use the attribute values to tell the GIS what colors and style to use when drawing features (see Fig.3.).
- The process of setting colors and drawing styles is often referred to as setting feature **symbolology**.

symbolology



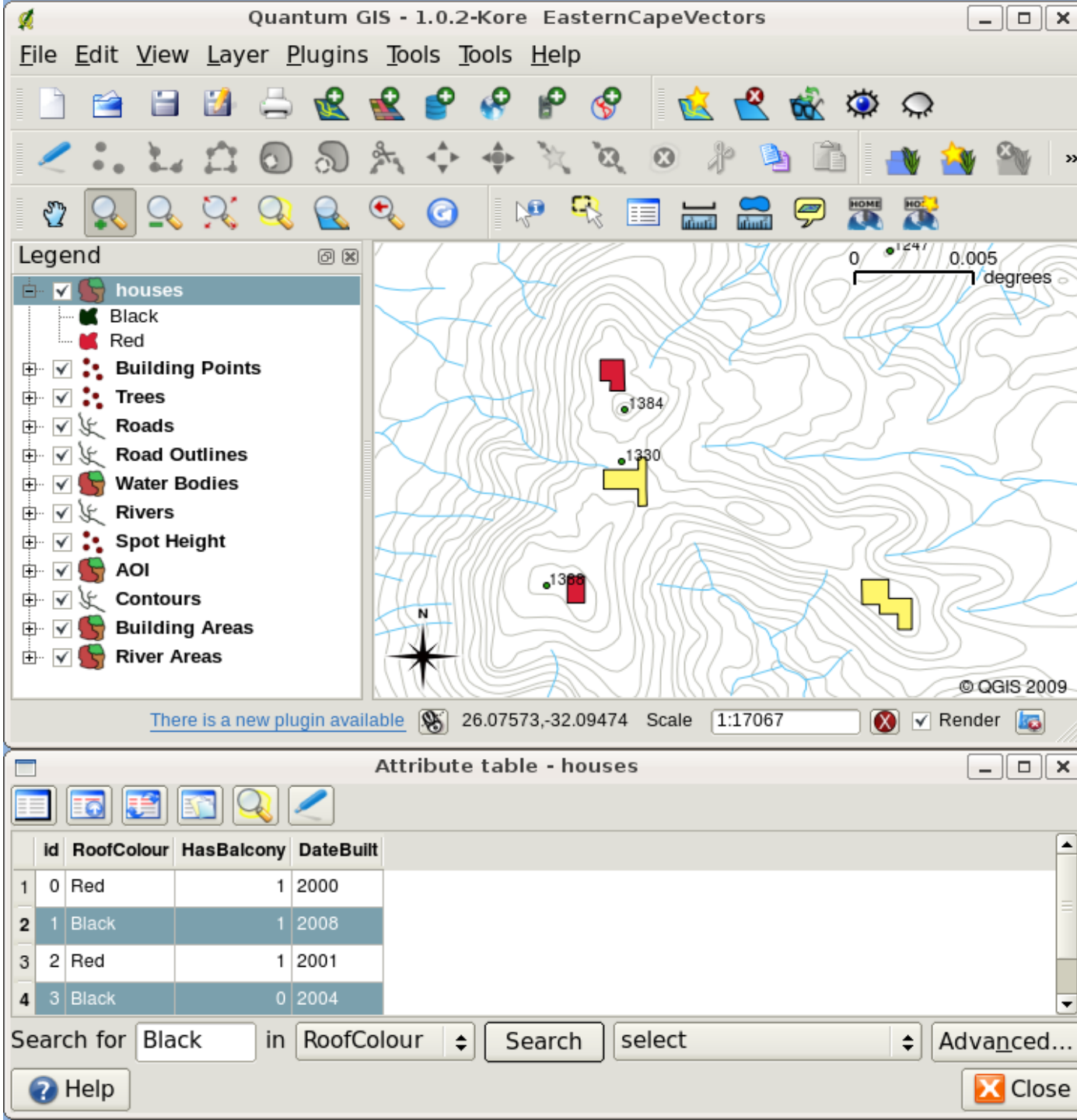
In a GIS Application,

we can draw features differently depending on their attributes.

- *On the **left** we have drawn house polygons with the same colour as the roof attribute.*
- *On the **right** we colour coded houses according to whether they have a balcony or not.*

Map labels.

- Attribute data can also be useful when creating **map labels**.
- Most GIS Applications will have a facility to select an attribute that should be used to label each feature.
- If you have ever **searched a map** for a place name or a specific feature, you will know how time consuming it can be. Having attribute data can make searching for a specific feature quick and easy.
- In Fig. 4. you can see an example of an attribute search in a GIS.



In a GIS Application,

- we can also search for features based on their attributes.*
- Here we see a search for houses with black roofs.*
- Results are shown in yellow in the map, and*
- Selected on the table*

- Finally, attribute data can be very useful in carrying out **spatial analysis**.
- **Spatial analysis** combines the **spatial information** stored in the geometry of features with their **attribute information**.
- This allows us to study features and how they relate to each other.
- There are many types of spatial analysis that can be carried out,

For example,

- you could use GIS to **find out how many red roofed houses** occur in a particular area.
- If you have **tree features**, you could use GIS to try to find out **which species might be affected if a piece of land is developed**.
- We can use the attributes stored for **water samples along a river course** to understand where **pollution is entering into the stream**.
- The possibilities are endless! In a later topic we will be exploring spatial analysis in more detail.

Before we move on to attribute data in more detail,
let's take a quick **Recap**.

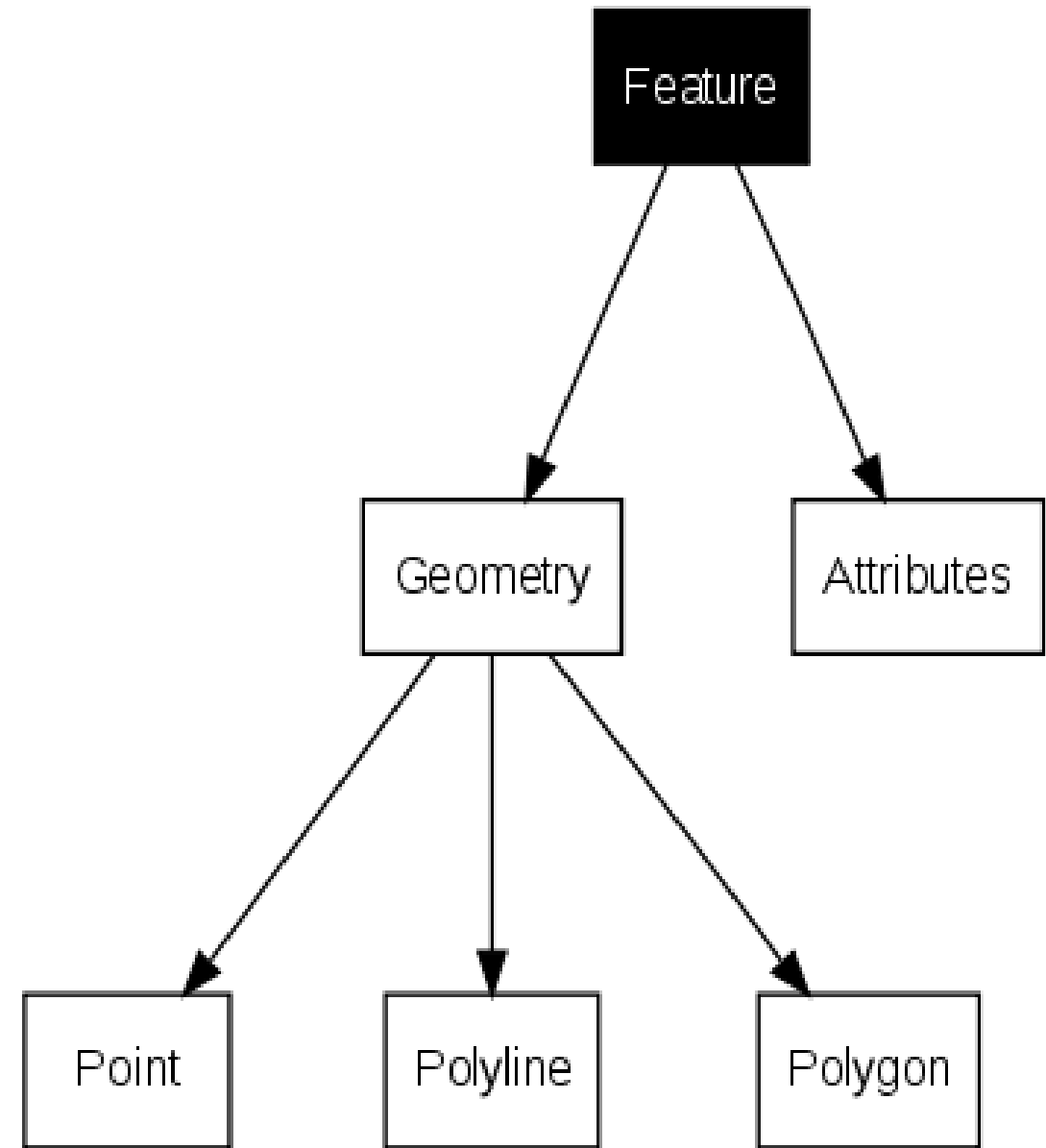
✓ Features are real world things such as

- Roads,
- Property boundaries,
- Electrical substation sites and
- so on.

✓ A **feature** has a **geometry** (determines if it is a **point**, **polyline** or **polygon**) and

✓ **attributes** (which describe the feature).

- This is shown in Fig.5.



Attributes in detail

- Attributes for a vector feature are stored in a **table**.
- A table is like a spreadsheet.
- ✓ Each column in the table is called a **field**.
- ✓ Each row in the table is a **record**.
- Table House Attribute shows a simple example of how an attribute table looks in a GIS.
- The **records** in the attribute table in a GIS each **correspond to one feature**.
- Usually the information in the attribute table is stored in some kind of database.
- The GIS application links the **attribute** records with the **feature geometry** so that **you can find records in the table by selecting features on the map**, and **find features on the map by selecting features in the table**.

Table House Attributes: An attribute table has fields (columns) and records (in rows).

Attribute Table	Field 1 : YearBuilt	Field 2: RoofColour	Field 3: Balcony
Record 1	1998	Red	Yes
Record 2	2000	Black	No
Record 3	2001	Silver	Yes

- Each field in the attribute table contains a specific type of data -- text, numeric or date.
- Deciding what attributes to use for a feature requires some thought and planning.
- In our house example earlier on in this topic, we choose roof color, presence of a balcony and month of construction as attributes of interest.

We could just as easily have chosen other aspects of a house such as:

- Number of levels
- Number of rooms
- Number of occupants
- Type of dwelling (RDP House, block of flats, shack, brick house, etc)
- Year the house was built
- Area of floor space in the house
- and so on....

- With so many options,
- how do we make a good choice as to what attributes are needed for a feature?
- It usually boils down to what you plan to do with the data.
- If you want to produce a color coded map showing houses by age, it will make sense to have a 'Year Built' attribute for your feature.
- If you know for sure you will never use this type of map, it is better to not store the information.
- Collecting and storing unneeded information is a bad idea because of the cost and time required to research and capture the information.
- Very often we obtain vector data from companies, friends or the government.
- In these cases it is usually not possible to request specific attributes and we have to make do with what we get.

Single Symbols

- If a feature is symbolized without using any attribute table data, it can only be drawn in a simple way.

For example

- with point features you can set the **color** and **marker** (circle, square, star etc.) but that is all.
- You cannot tell the GIS to draw the features based on one of its properties in the attribute table.
- In order to do that, you need to use either **graduated**, **continuous** or **unique value** symbol.
- These are described in detail in the sections that follow.

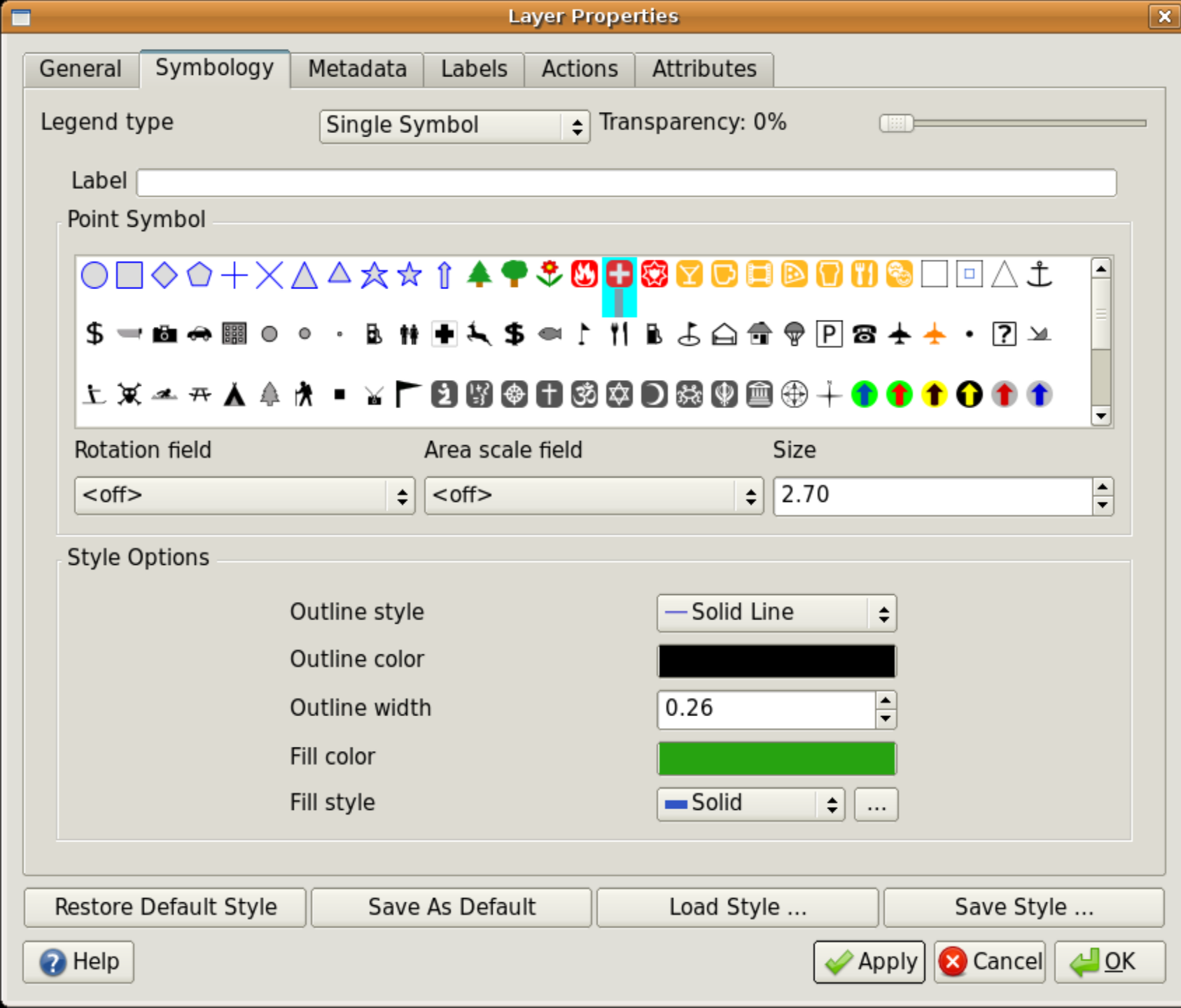


Fig.6.

- When using simple symbols, the feature is drawn without using an attribute to control how it looks.
- This is the dialog for point features.

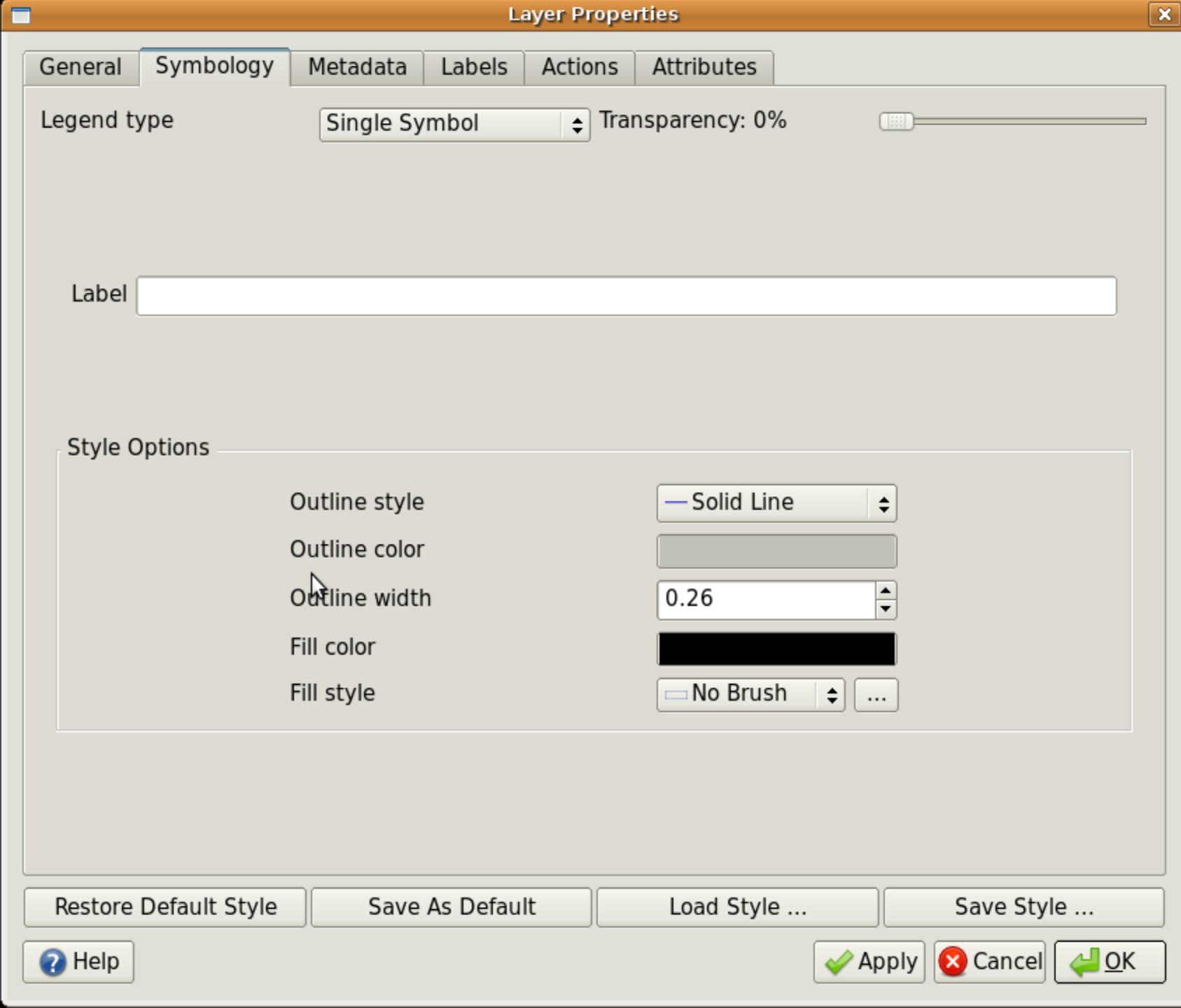


Fig. 7.

There are different options when defining simple symbols for polyline and polygon features.

Single Symbols

- A GIS application will normally allow you to set the symbology of a layer using a **dialog box** such as the one shown in in Fig. 6.
- In this dialog box you can choose colors and symbol styles.
- Depending on the geometry type of a layer, different options may be shown.
- For example with point layers you can choose a **marker style**.
- With line and polygon layers there is no marker style option, but instead you can select a **line style** and **color** such as dashed orange for gravel roads, solid orange for minor roads, and so on (as shown in Fig. 7.).
- With polygon layers you also have the option of setting a **fill style** and color.

Graduated Symbols

- Sometimes vector features represent things with a changing numerical value.
- Contour lines are a good example of this.
- Each contour usually has an attribute value called 'height' that contains information about what height that contour represents.
- In earlier in this topic we showed contours all drawn with the same colour.
- Adding colour to the contours can help us to interpret the meanings of contours.

For example we can draw

- ✓ low lying areas with one color,
- ✓ mid-altitude areas with another and
- ✓ high-altitude areas with a third.

Layer Properties

General **Symbology** Metadata Labels Actions Attributes

Legend type: Graduated Symbol Transparency: 0%

Classification field: HEIGHT

Mode: Quantiles

Number of classes: 3

Classify Delete class

980.000000 - 1120.000000
1120.000000 - 1240.000000
1240.000000 - 1500.000000

Label

Style Options

Outline style: Solid Line

Outline color: [Purple]

Outline width: 0.26

Fill color: [Black]

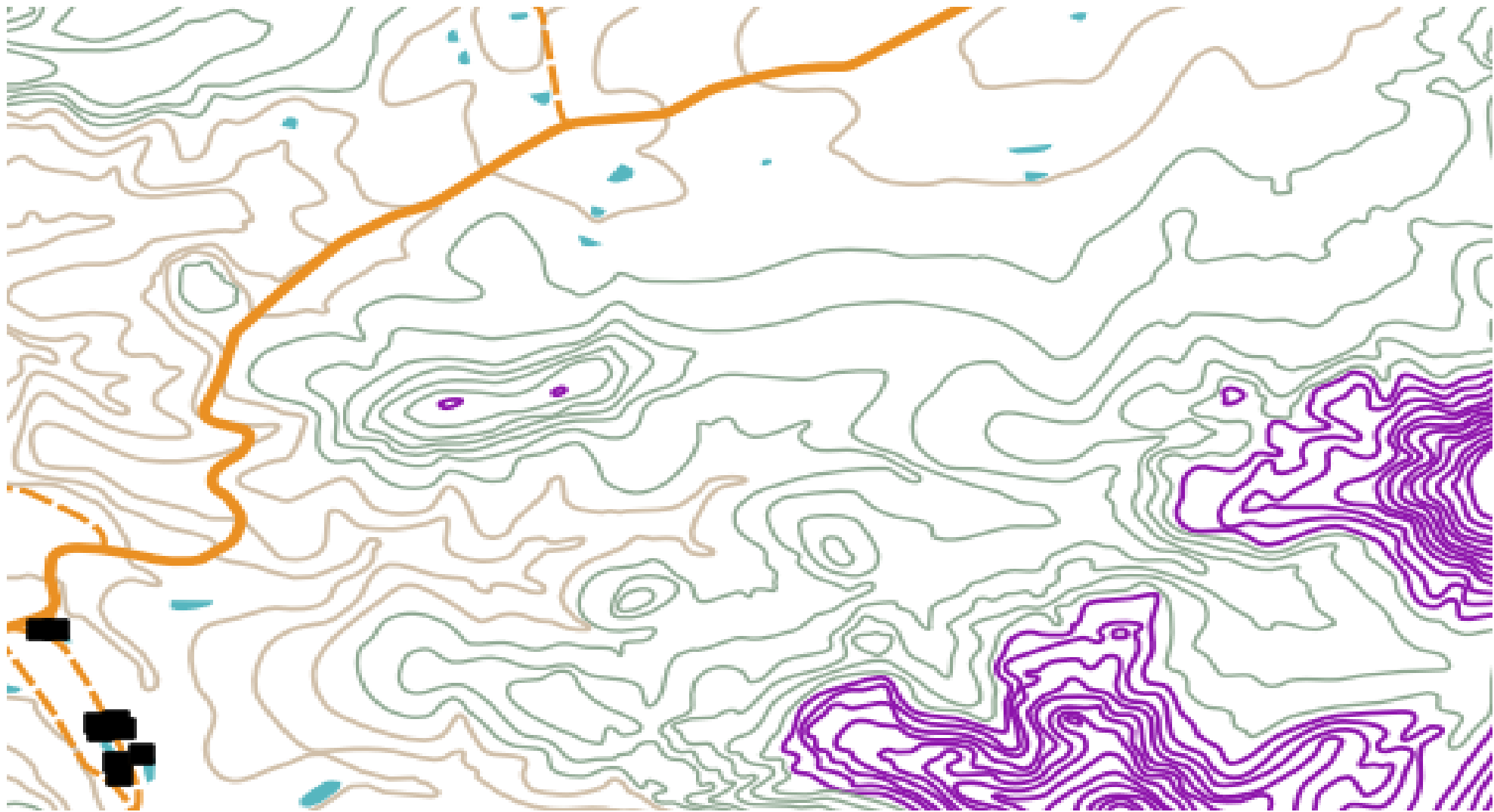
Fill style: Solid

Restore Default Style Save As Default Load Style ... Save Style ...

? Help Apply Cancel OK

The height attribute of contours can be used to separate the contours into 3 classes.

- *Contours between 980 m and 1120 m will be drawn in brown,*
- *those between 1120 m and 1240 m in green and*
- *those between 1240 m and 1500 m in purple.*



Our map after setting graduated colors for our contours.

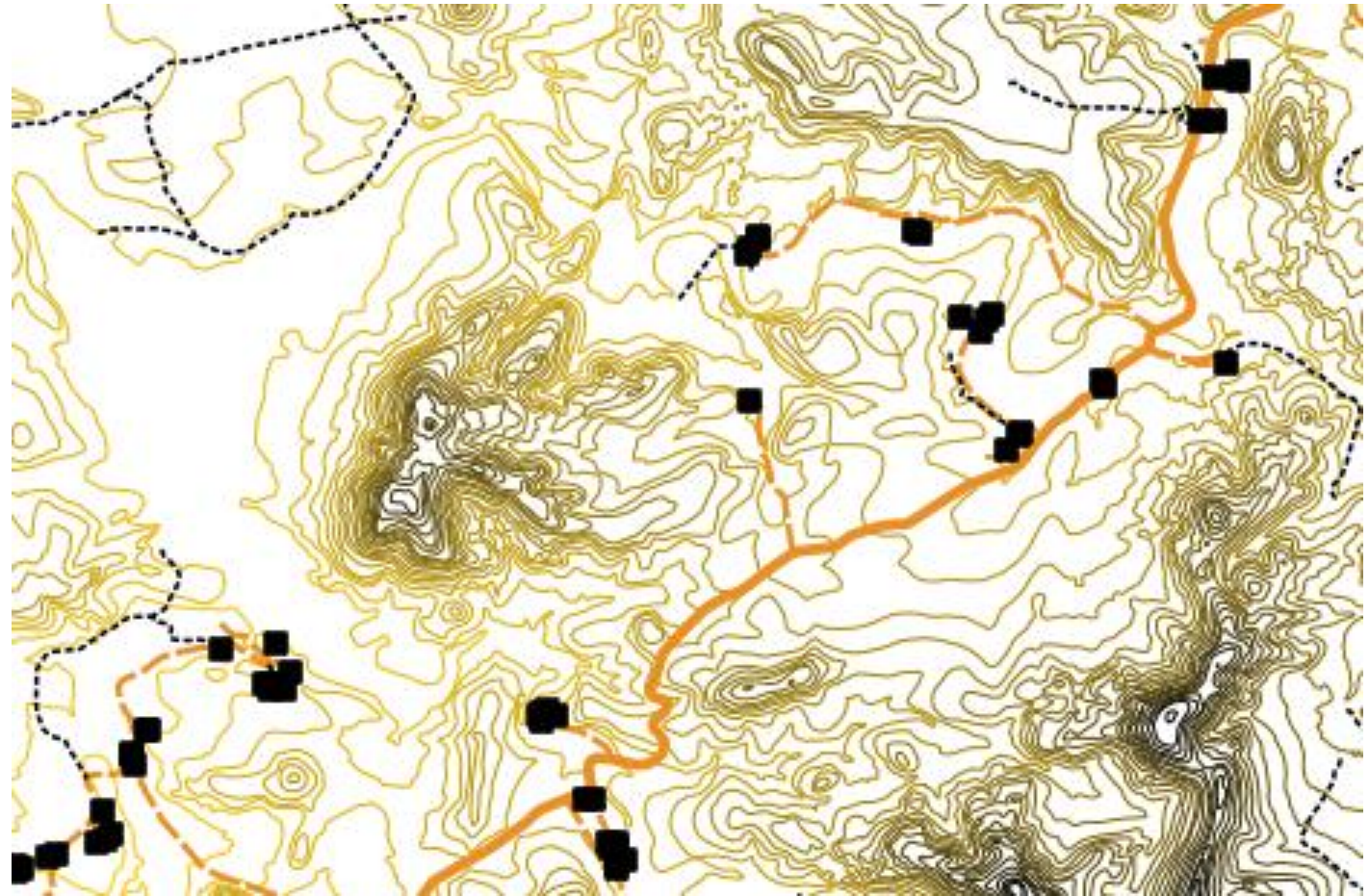
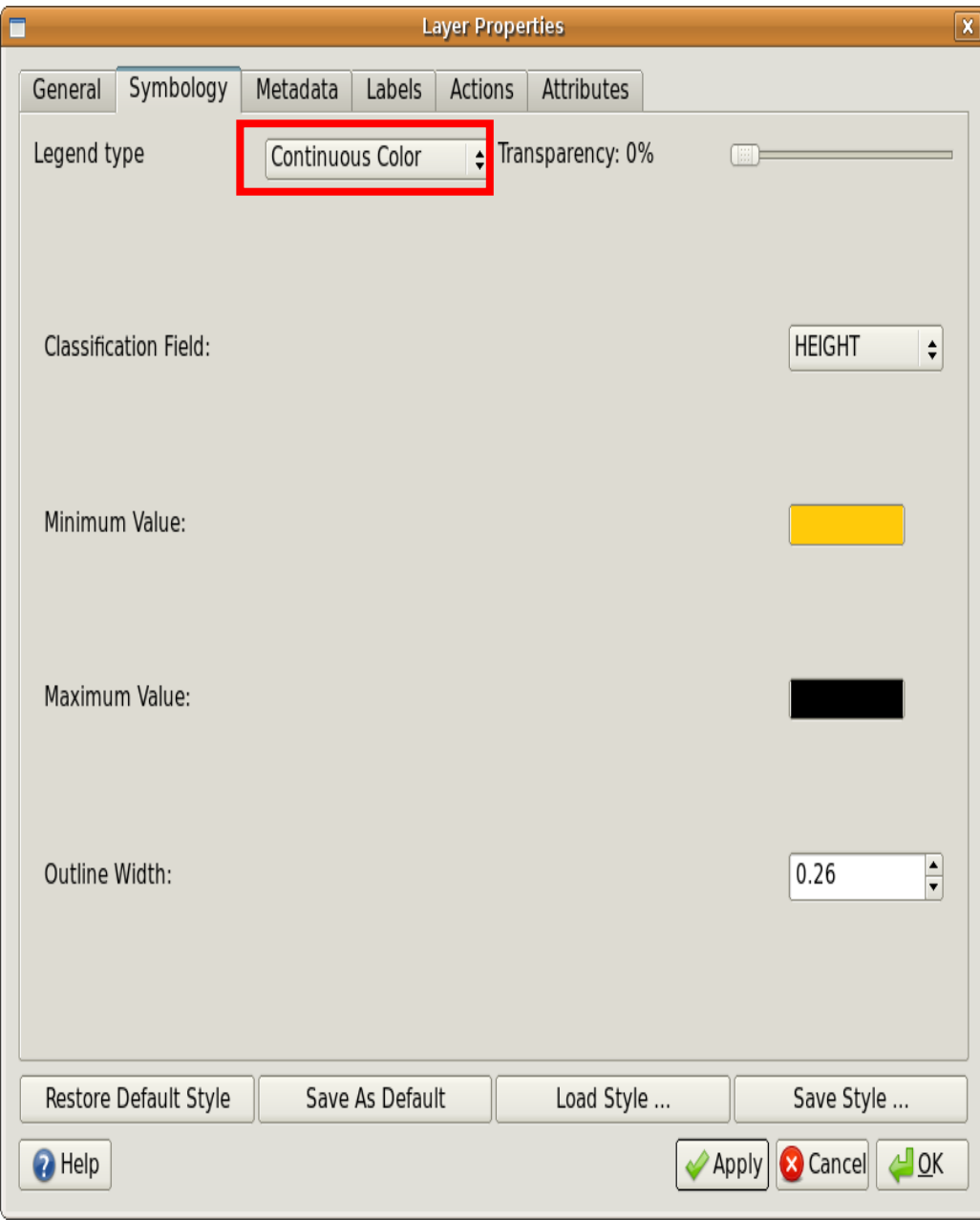
Graduated Symbols

- **Graduated symbols** are most useful when you want to show clear differences between features with attribute values in different value ranges.
- The GIS Application will analyze the attribute data (e.g. height) and, based on the number of classes you request, create groupings for you.
- This process is illustrated in Table_graduated.

Attribute Value	Class and Colour
1	Class 1
2	Class 1
3	Class 1
4	Class 2
5	Class 2
6	Class 2
7	Class 3
8	Class 3
9	Class 3

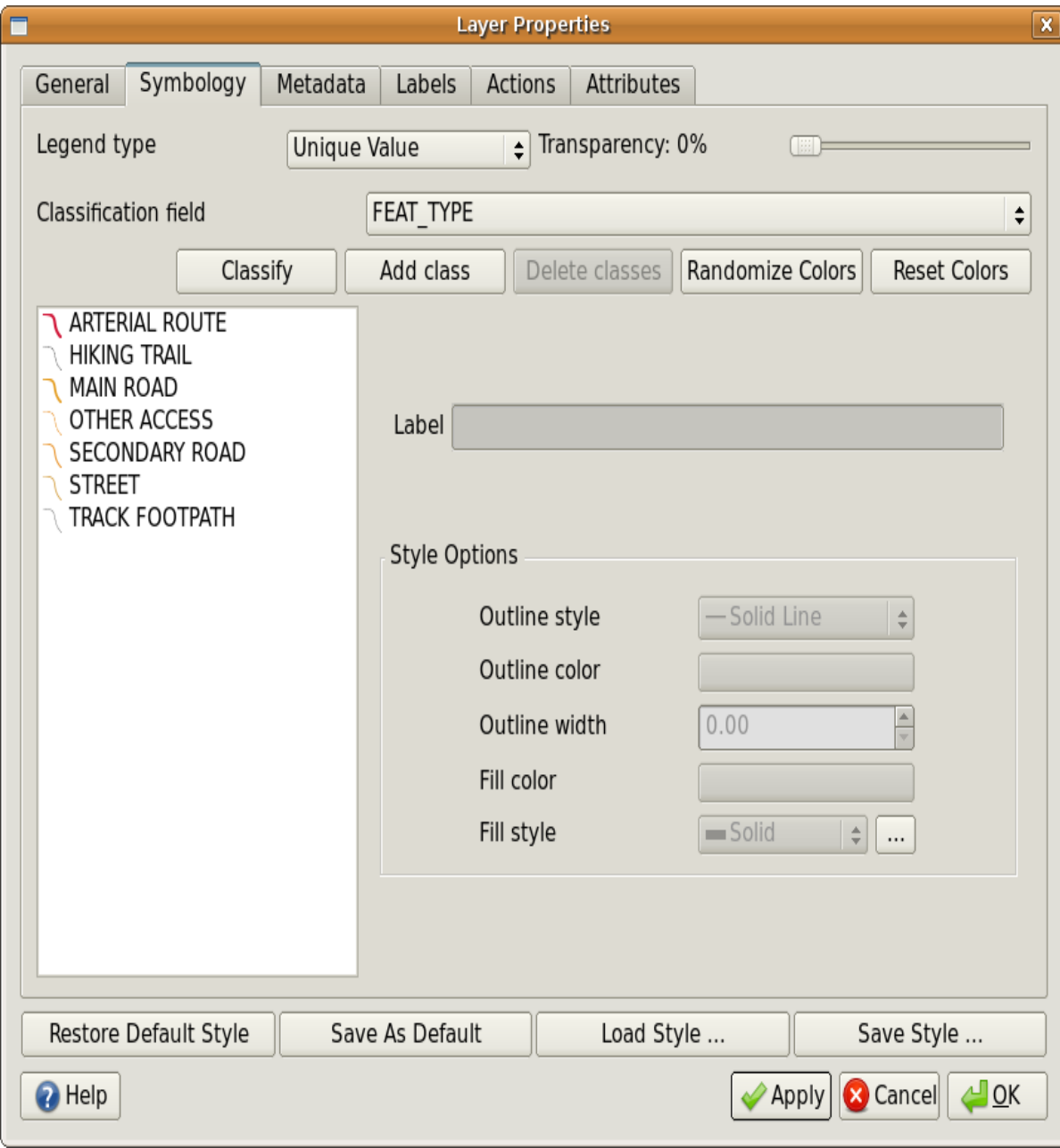
Table_graduated.

Continuous Color Symbols



A contour map drawn using continuous color symbology

Unique Value Symbols



A roads vector layer symbolized using a unique value per road type.

Things to be aware of:

Deciding which attributes and symbology to use requires some planning:

- Before you start collecting any **GeoSpatial** data, you should ensure you know
 - what attributes are needed and
 - how it will be symbolized.
- ❖ It is very difficult to go back and re-collect data if you plan poorly the first time around.
- ❖ Remember also that the goal of collecting attribute data is to allow you to analyze and interpret spatial information.
- ❖ How you do this depends on the questions you are trying to answer.
- ❖ **Symbology is a visual language** that allows people **to see and understand your attribute** data based on the **colors and symbols** you use.
- ❖ Because of this you should put a lot of thought into how you symbolize your maps in order to make them easy to understand.

What have we learned?

- Let's wrap up what we covered in this worksheet:
- Vector features have **attributes**
- Attributes **describe** the **properties** of the feature
- The attributes are stored in a **table**
- Rows in the table are called **records**
- There is **one record per feature** in the vector layer
- Columns in the table are called **fields**
- Fields represent **properties** of the feature e.g. height, roof color etc.
- Fields can contain **numerical**, **string** (any text) and **date** information
- The attribute data for a feature can be used to determine how it is **symbolized**.
- **Graduated color** symbology groups the data into discrete classes
- **Continuous color** symbology assigns colors from a color range to the features based on their attributes
- **Unique value** symbology associates each different value in the chosen attribute column with a different symbol (color and style)
- If the attribute of a vector layer is not used to determine its symbology, it is drawn using a **single symbol** only

Now you try!

Here are some ideas for you to try :

Using the table that you created in the last topic,

- add a new column for the symbology type you would use for each feature type and identify which symbology type they would use (see [table_example_symbols](#) for an example).
- Try to identify which symbology types you would use for the following types of vector features:
 - points showing [pH level of soil samples](#) taken around your school
 - lines showing a [road network](#) in a city
 - polygons for [houses](#) with an attribute that shows whether it is made of brick, wood or 'other' material.

Real world feature	Geometry Type	Symbology Type
The school flagpole	Point	Single Symbol
The soccer field	Polygon	Single Symbol
The footpaths in and around the school	Polyline	Have your learners count the number of learners using each footpath in the hour before school and then use graduated symbols to show the popularity of each footpath
Places where taps are located	Point	Single symbol
Classrooms	Polygon	Unique value based on the grade of the learners in the classroom
Fence	Polyline	Have your learners rate the condition of the fence around your school by separating it into sections and grading each section on a scale of 1-9 based on its condition. Use graduated symbols to classify the condition attribute.
Classrooms	Polygon	Count the number of learners in each classroom and use a continuous colour symbol to define a range of colours from red to blue.

Table Example Symbols:
 An example of a table that defines the feature types and the kind of symbology you would use for each.

- **Something to think about**
- If you don't have a computer available, you can use transparency sheets and a 1:50 000 map sheet to experiment with different symbology types.

For example:

- Place a transparency sheet over the map and using different colored pens, draw in **red** all contour lines below 900 m (or similar) and in **green** all lines above or equal to 900 m.
- Can you think of how to reproduce other symbology types using the same technique?

- Further reading
- Website: <https://en.wikipedia.org/wiki/Cartography>
- What's next?
- In the section that follows we will take a closer look at **data capture**. We will put the things we have learned about vector data and attributes into practice by creating new data.